

Quasi-isospectrality of manifolds and Schrödinger Operators

(with comments) (talk in Göttingen, Pseudodifferential Techniques in Singular Analysis)

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I. Isospectrality & Quasi-isospectrality

- Isospectral problem: If two operators have the same spectrum, are they equal (or equivalent)? To which extent is an operator determined by its spectrum?
- In differential geometry: let (M, g) be a n -dim^l Riemannian mfd

Δ_g : Laplacian on functions

$\Delta_g \psi = \lambda \psi$, $\psi \neq 0$, the eigen value problem

$$\text{Spec}(\Delta_g) = \{\lambda_j, j \geq 0\}$$

[Q: $\text{Spec}(\Delta_g) = \text{Spec}(\Delta_h) \Rightarrow \exists \psi: M \rightarrow M : h = \psi^* g$?

Question: To which extent is the metric g determined by $\text{Spec}(\Delta_g)$

(Large amount of literature about the general problem, as many of you probably already know)

Question: Could one relax the conditions of isospectrality and still obtain some of the known results of isospectrality, e.g. existence and compactness of isospectral sets?

(After looking at a couple of options, we decided to study the concept of isospectrality).

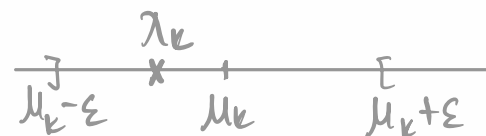
Defⁿ. (Two closed self-adjoint operators are quasi-isospectral if their spectra are the same with exception of one (or finitely many) eigenvalues that is allowed to vary in a prescribed interval.)

let T_1, T_2 be closed, non-negative s.a. operators with discrete spectrum. $\text{Spec}(T_1) = \{\lambda_j\}_{j \geq 0}$, $\text{Spec}(T_2) = \{\mu_j\}_{j \geq 0}$

let $\varepsilon > 0$ and $k \geq 1$, we say that T_1 and T_2 are (k, ε) -quasi-isospectral if:

i) $\lambda_m = \mu_m$ for $m \neq k$

ii) $\lambda_k \in]\mu_k - \varepsilon, \mu_k + \varepsilon[$, and



iii) $\lambda_m, \mu_m \notin]\mu_k - \varepsilon, \mu_k + \varepsilon[$ if $m \neq k$

* As presented in the talk, the relation is not symmetric. Therefore an extra condition should be added to iii)

$\lambda_m, \mu_m \notin]\lambda_k - \varepsilon, \lambda_k + \varepsilon[$ if $m \neq k$.

(Thanks for pointing this out).

II. Quasi-isospectral Riemannian manifolds.

M^n : closed manifold, g a fixed Riemannian metric on M .

Recall the asymptotic expansion of the heat trace

$$T_n(e^{-t\Delta_g}) = \sum_{j=0}^{\infty} e^{-t\lambda_j} \sim t^{-n/2} \sum_{j=0}^{\infty} a_j(t) t^j \quad \text{as } t \downarrow 0$$

(as in the talks by Julie Rowlett on flat tori, and Gustav Mårdby on alcoves).

Thm 1 (C.A & C. Perez)

Let h be (ϵ, ϵ) -quasi-isospectral to g . Then:

i) If n is odd then h and g are isospectral.

ii) If n is even, then

- $a_j(h) = a_j(g)$ for $j < 1 + \frac{n}{2}$

- $|a_{1+\frac{n}{2}}(h) - a_{1+\frac{n}{2}}(g)| \leq \epsilon$, and

- $\exists K = K(\lambda_i(g), \epsilon) > 0$ such that

$$|a_j(h) - a_j(g)| \leq \epsilon \quad \forall j > K \quad \text{uniformly in } j$$

Proof (Very simple & straight forward, but still nice)

$$\lambda_m(h) = \lambda_m(g) + s \delta_{m,i} \quad \text{for some } s \in]-\epsilon, \epsilon[$$

we know that $e^{-t\Delta_g}$, $e^{-t\Delta_h}$ and $e^{-t\Delta_h} - e^{-t\Delta_g}$ are trace class and their traces have asymptotic expansions of Poincaré type w.r.t. the sequence $(t^{-n/2+j})_{j \geq 0}$ as $t \rightarrow 0^+$ for $t \in]0, \infty[$.

By the theory of asymptotic expansions, we have that if $\psi_1(t) \sim \sum c_j \psi_j(t)$, $\psi_2(t) \sim \sum a_j \psi_j(t)$, and $(\psi_1 + \psi_2)(t) \sim \sum b_j \psi_j(t)$ as $t \rightarrow t_0$ then $b_j = c_j + a_j \quad \forall j$

$$\text{let } F(t) = \text{Tr}(e^{-t\Delta_h}) - \text{Tr}(e^{-t\Delta_g}) = e^{-t\lambda_i(g)} (e^{-ts} - 1)$$

choosing the right inequalities we can prove that

$$|F(t)| \leq 2e^{-t\lambda_i(g)} \epsilon t \Rightarrow F(t) = O(t) \text{ as } t \searrow 0$$

On the other hand

$$F(t) = t^{-n/2} \sum_{j=0}^{\infty} b_j(h, g) t^j \Rightarrow b_j = 0 \text{ for } j < 1 + \frac{n}{2} \Rightarrow a_j(h) = a_j(g)$$

Now, $F(t) = \sum_{j=0}^{\infty} e^{-t\lambda_j(h)} - \sum_{j=0}^{\infty} e^{-t\lambda_j(g)} = e^{-t(\lambda_i(g)+s)} - e^{-t\lambda_i(g)}$ } entire function on $t \dots$

$$\Rightarrow \underbrace{t^{-j/2} \sum_{j=0}^{\infty} b_j(h, g) t^j}_{\text{if } n \text{ odd, only half-integer powers}} = \underbrace{\sum_{l=0}^{\infty} (-1)^l \frac{(\lambda_i+s)^l - \lambda_i^l}{l!} t^l}_{\text{integer powers}}$$

• If n is odd $\Rightarrow b_j(h_1, h_2) = 0 \quad \forall j \geq \lceil 1 + \frac{n}{2} \rceil$

$\Rightarrow (\lambda_i+s)^l - \lambda_i^l = 0 \quad \text{for } l \geq 1 \Rightarrow s=0 \Rightarrow h, g \text{ isospectral}$

• If n is even $a_j(h) - a_j(g) = (-1)^l \frac{(\lambda_i+s)^l - \lambda_i^l}{l!} \quad \text{for } l = j - \frac{n}{2} \geq 1$

$\Rightarrow l=1 \Rightarrow a_{1+\frac{n}{2}}(h) - a_{1+\frac{n}{2}}(g) = -s$

For $l > 1$ apply mean value theorem to $(\lambda_i+s)^l$ (in s)

$$|a_j(h) - a_j(g)| \leq \underbrace{\frac{(\lambda_i+\varepsilon)^{l-1}}{(l-1)!}}_{=: \ell(l-1)} 2\varepsilon$$

as function of $l \rightarrow \ell(x) = \frac{(\lambda_i+\varepsilon)^x}{\Gamma(x+1)}$

$\ell(x) \rightarrow 0 \quad \text{as } x \rightarrow \infty$ and ℓ has a maximum

$\Rightarrow \exists K > 0 \text{ s.t. } \forall l \geq K \quad \ell(l-1) \leq 1$ ■

For manifolds with boundary we have:

Thm 2 (C.A & C. Perez) Let N^n mfd w. b.dary. $\partial N \neq \emptyset$. Dirichlet Laplacian, here $\text{Tr}(e^{-t\Delta_{N,0}}) \sim t^{-n/2} \sum_{j=0}^{\infty} a_j t^{j/2}$. If h and g are (i, ε) -quasi-isospectral

then:

- 1) $a_j(h) = a_j(g)$ if $j \leq n+1$ or $\left. \begin{array}{l} \text{if } n \text{ even and } j \text{ odd} \\ \text{if } n \text{ odd and } j \text{ even.} \end{array} \right\}$

2) $|a_{n+2}(h) - a_{n+2}(g)| \leq 2\varepsilon$

3) $\exists K = K(i, \varepsilon) : \forall l \geq K \quad |a_l(h) - a_l(g)| \leq 2\varepsilon$.

Rmk Notice that the same holds if we perturb finitely many eigenvalues instead of only one.

III Schrödinger Operators on manifolds

(Back to the motivation: To extend known compactness results)

(M^n, g) closed Riemannian manifold, $V \in C^\infty(M)$

$$I_S(V) = \{W \in C^\infty(M) \mid \text{Spec}(\Delta_g + V) = \text{Spec}(\Delta_g + W)\}$$

[Thm (Bruning '84) If $n \leq 3$ $I_S(V)$ is compact in $C^\infty(M)$

(Donnelly 2004) If $n \leq 9$ $I_S^+(V) = \{W \in I_S(V) \mid W \geq 0\}$ is compact in $C^\infty(M)$

The idea of the method of proof goes back to R. Melrose's "Isospectral sets of drumheads are compact in C^∞ ". This is:

Isospectral $\Rightarrow$ same heat invariants $\Rightarrow$ explicit form of the h.i. $\Rightarrow$ uniform estimates on $\|V\|_{S,2}$

$a_j(g, V)$

$\int_M P_j(g, V) d\mu$

using bootstrapping argument

universal polynomials in the covariant derivatives of V and of the curvature tensor of g .

$\Rightarrow$ Compactness.

* Bruning: By the { Sobolev and Gagliardo-Nirenberg } inequalities, a uniform estimate in $W_{3n_0-2,2}(M)$, $n_0 = \lceil \frac{n}{2} \rceil$ is enough to obtain compactness if $n \leq 3$.

For $V \in C^\infty(M)$ define

$$QIS_{K,\varepsilon}(V) = \{W \in C^\infty(M) \mid \Delta + W \text{ and } \Delta + V \text{ are } (K, \varepsilon)\text{-quasi-isospectral}\}$$

Thm 1 and Thm 2 extend to this setting as well as the compactness results by Bruning and Donnelly mentioned above.

Now, we look in particular to Schrödinger operators on an interval where there exist quasi-isospectral potentials.

IV. Schrödinger Operators on an interval - Existence

Explicit construction by Bilotta, Morassi & Turco (1993)

Method

- also found in the book by Pöschel-Traubowitz (1987)
- Based on * the single commutation method
* the double commutation method

developed by Gel'fand, Levitan, Darboux, Crum, Deift,
Gesztesy-Telsch,

to insert and remove an eigenvalue in a spectral gap.

Let $q \in L^2[0,1]$, $\mathcal{Z}_q$: operator associated to the problem

$$\begin{cases} -y'' + qy = \lambda y & x \in (0,1) \\ y(0) = 0 = y(1) \end{cases}$$

Defⁿ. Two potentials $q, p \in L^2[0,1]$ are quasi-isospectral if the operator $\mathcal{Z}_q$ and $\mathcal{Z}_p$ are.

Existence: Given q , construct p quasi-isospectral to q .

Statement : (BMT, PT): let $q \in L^2[0,1]$, $\text{Spec}(\mathcal{Z}_q) = \{\lambda_k(q)\}_{k \geq 1}$
set $\lambda_0 = 0$, let $t \in \mathbb{R}$, $n \geq 1$ be such that
$$\lambda_{n-1}(q) < \lambda_n(q) + t < \lambda_{n+1}(q)$$

Then $\exists p \in L^2[0,1]$: $\lambda_m(p) = \lambda_m(q) + t \delta_{m,n} \quad \forall m \geq 1$

In order to build p we need Darboux's lemma

Lemma (Darboux) let $\mu \in \mathbb{R}$ be fixed, and

let $f \neq 0$ s.th. $-f'' + qf = \lambda f$, $\lambda \neq \mu$

let $g \neq 0$ s.th. $-g'' + qg = \mu g$

then $h = \frac{1}{g} (gf' - g'f)$ satisfies

$$-h'' + (q - 2 \frac{d^2}{dx^2} \log(g)) h = \lambda h \quad (*)$$

Moreover, if $\lambda = \mu$ the general solution of $(*)$ is

$$\frac{1}{g} (a + b \int_0^x g^2(s) ds) \quad \text{for } a, b \in \mathbb{R}$$

Construction of p (simplified)

Consider $\{\ell_m\}_{m \geq 1}$ $- \ell_m'' + q \ell_m = \lambda_m \ell_m$ $\ell_m(0) = 0$
 $\ell_m'(0) = 1$

let n and t be as above, consider

$y_1 = y_1(x, q, \lambda+t)$: $-y_1'' + q y_1 = (\lambda+t) y_1$
 $y_1(0) = 1, y_1'(0) = 0$

constructs w_t that satisfies

$-w_t'' + q w_t = (\lambda+t) w_t$
 $w_t(0) = 1$ and $w_t(1) = y_1(1, q, \lambda)$

Now there are 3 Steps

Step 1 DL with $f = w_t$, $g = \ell_n$, $B_{n,t} = w_t \ell_n' - w_t' \ell_n$

then $h = \frac{B_{n,t}}{\ell_n}$ satisfies $-h'' + \hat{q} h = (\lambda+t) h$

Step 2 DL with $f = \ell_m$ and $g = \ell_n$

then e : $-e'' + \hat{q} e = \lambda_m e$

$$(\hat{q} = q - 2 \frac{d^2}{dx^2} \log(\ell_n))$$

Step 3 DL with $f = \ell$ and $g = h$

$\Rightarrow K_{m,t}$ that satisfies $-K_{m,t}'' + p K_{m,t} = \lambda_m K_{m,t}$

with $p = q - 2 \frac{d^2}{dx^2} \log(\beta_{n,t})$

if $m=n$ $K_{n,t} = \frac{\ell_n}{\beta_{n,t}}$

To notice: $\beta_{n,t} > 0$ & $\beta_{n,t} \in L^2$.

$K_{m,n}$ are well defined, non-trivial and satisfy the bdcy condition.

Example $q=0$, $-y'' = \lambda y$ $y(0)=0$ $y(1)=0$ $\text{Spec}(\mathcal{L}_0) = \text{Spec}(\Delta) = \{n^2\pi^2\}_{n \geq 1}$

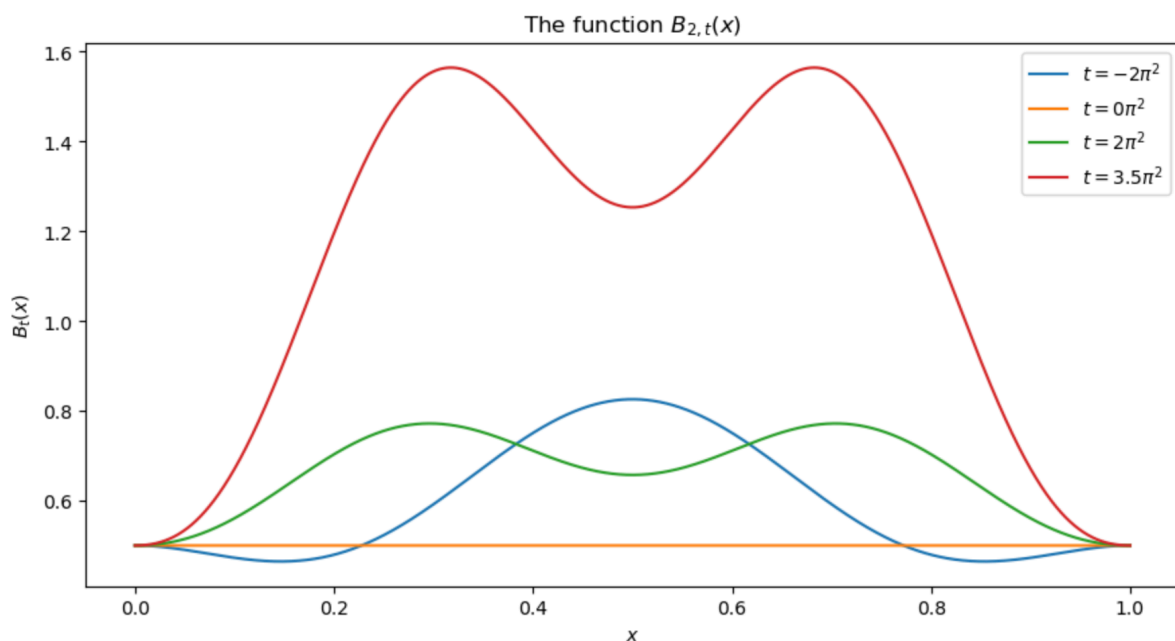
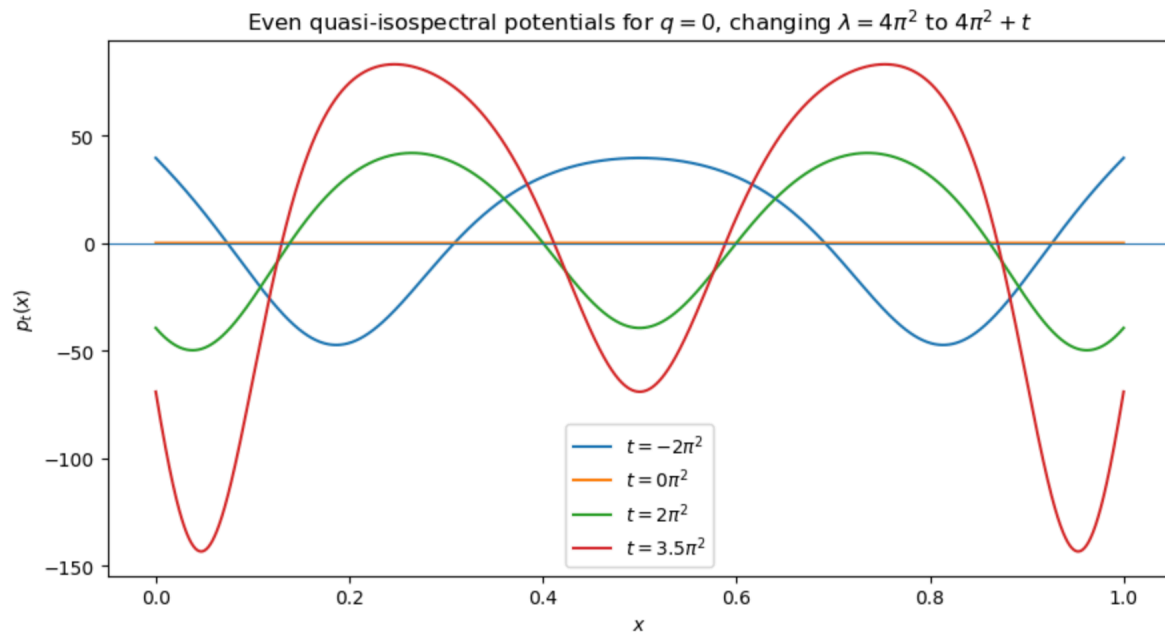
$n \geq 2$ $(n-1)^2 \pi^2 < \mu < (n+1)^2 \pi^2$, $\mu = n^2 \pi^2 + t$

Using the method we obtain a family of even potentials that are quasi-isospectral to $q=0$ $\hookrightarrow$ respect to $x=1/2$

$p_{n,t} = - \frac{d^2}{dx^2} \log(\beta_{n,t})$

$$\beta_{n,t} = \frac{1}{\sin \sqrt{\mu}} \left(\cos(n\pi x) \left(\sin(\sqrt{\mu}(1-x)) + (-1)^n \sin(\sqrt{\mu}x) \right) + \frac{\sqrt{\mu}}{n\pi} \sin(n\pi x) \left(\cos(\sqrt{\mu}(1-x)) + (-1)^n \cos(\mu x) \right) \right)$$
 } even

Graphics made in Python with assistance of ChatGPT



Notice that some of these values of t plotted in these graphics don't satisfy the additional symmetry condition added to iii) in the definition of being (R, ε) -quasi-isospectral, but still give a 2-quasi-isospectral potential if the ε -part of the condition is relaxed to the existence of let us say, a pair $\varepsilon, \varepsilon'$.